The Illusion of Precision in Quantum Mechanics

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Abstract

A fitted model is exact in the aspect it was fitted to. That exactness is automatic and carries no information, because it was arranged; what a model does *outside* its fit is the only thing that can fail, and therefore the only thing that is evidence. Textbook quantum mechanics begins by reporting the hydrogen ground state as -0.500000 and treating this as exact, then concludes that the Schrödinger equation is exact for any number of electrons and merely uncomputable. Both steps deserve examination. The figure -0.500000 mixes a prediction with a convention: the pattern $1/n^2$ is predicted, while the coefficient $\frac{1}{2}$ defines atomic units, and the bridge from atomic energy to measurable frequency is Planck's constant, itself fixed by the spectra it is then used to reproduce. The bound-state spectrum is thus the fitted aspect, and exactness there is free. The first genuinely unfitted predictions fail: applied correctly as a two-body problem, which it handles exactly, the equation reaches only 1.1×10^{-5} on the ground state, and it makes $2s$ and $2p$ degenerate against a measured splitting of 1058 MHz. It is the leading term of an expansion completed by Dirac and by quantum electrodynamics. The $3N$-dimensional equation for N electrons is then a separate postulate, never solved without approximation beyond three or four electrons, so its exactness is supported only by extrapolation from a case where exactness is tautological. The electron's anomalous moment to twelve digits, usually offered in evidence, is a one-particle vacuum quantity of quantum electrodynamics and says nothing about it. We set against this a model of non-overlapping electron charge densities in three dimensions — equally a postulate, with no claim to exactness — whose energies, geometries and ionization energies come out within a few per cent from one electron to the ten thousand of a protein, parameter-free and computable. The comparison is not accuracy against inaccuracy, but a precision confined to few-electron systems against an accuracy that can be stated at every size.

1 The premise, checked

The claim that organises every introduction to the subject is that the Schrödinger equation solves hydrogen exactly. Written for one electron about a fixed point charge,

$$-\tfrac{1}{2}\nabla^2\psi - \frac{1}{r}\,\psi = E\psi, \tag{1}$$

it has the ground state $E = -\frac{1}{2}$ in atomic units, which is 13.605693 eV, and a spectrum $E_n = -1/2n^2$ reproducing the Balmer series. The agreement was decisive in 1926 and remains the reason the equation is believed.

It is not, however, exact, and the inexactness must be identified carefully, because one candidate explanation is not a defect of the equation at all. The nucleus moves, and (1) holds it fixed — but that is a shortcoming of how the equation has been *applied*, not of the equation. Written for two bodies it separates exactly into centre-of-mass and relative motion, and the relative part returns the hydrogenic spectrum with the reduced mass $\mu = m_e m_p/(m_e + m_p)$ in place of m_e, giving 13.598287 eV. That is the equation's own answer, and it is obtained exactly.

The residual against the measured 13.598435 eV [1] is

$$0.000147 \text{ eV}, \qquad 1.1 \times 10^{-5} \text{ relative},$$

and this one is a genuine deficiency: it is the relativistic and radiative physics the equation does not contain, supplied by the Dirac equation and by quantum electrodynamics.

The spectrum fails more sharply than the energy, and qualitatively rather than numerically. Equation (1) makes $2s$ and $2p$ exactly degenerate. They are not: the splitting is 1057.845 MHz, the Lamb shift [2], whose discovery in 1947 initiated renormalised quantum electrodynamics. Fine structure enters at order $\alpha^2 \approx 5 \times 10^{-5}$ of the binding energy and the Lamb shift at order $\alpha^3 \approx 4 \times 10^{-7}$; the Schrödinger equation contains neither.

So the equation that is said to be exact for hydrogen is the leading term of an expansion. Its accuracy for the ground-state energy is five significant figures, not infinitely many, and it predicts a degeneracy that does not exist. This is well known in atomic physics and is not a criticism of the equation. It is a correction to what the equation is taken to establish.

2 Where the form came from

Equation (1) has two ingredients. The potential term is the Coulomb interaction, taken from electrostatics and therefore empirical. The kinetic term is the quadratic form

$$T[\psi] \;=\; \tfrac{1}{2} \int |\nabla \psi|^2 \, dx, \tag{2}$$

and this is a choice. It was not derived from anything prior; Schrödinger arrived at it by analogy with the Hamilton–Jacobi theory and de Broglie's relation, and it was adopted because the resulting eigenvalues matched the hydrogen spectrum. That is model selection by agreement with measurement, which is the ordinary way physics proceeds and is not a defect.

It is worth separating what is predicted here from what is arranged, because the famous -0.500000 consists of both.

The physical content is the *pattern* $E_n \propto 1/n^2$. That is a prediction, it is not trivial, and it is what made the equation believable in 1926.

The *number* is not. The coefficient $\frac{1}{2}$ in (2), and with it a ground state at exactly $-\frac{1}{2}$, is a statement about units: atomic units are defined so that the hydrogen ground state comes out $-\frac{1}{2}$, which makes that figure a convention rather than an outcome. Converting it into anything measurable requires a bridge from atomic energy to the energy of light, and that bridge is Planck's constant, through $\Delta E = h\nu$. But h — equivalently R_∞, since the Hartree is $2R_\infty hc$ — is itself fixed by spectroscopy, and the hydrogen spectrum is among the measurements that fix it. The scale is read off the spectrum and then put back into it.

Taken together: the choice of (2) among the quadratic forms one might write, and the constant connecting its eigenvalues to observed frequencies, were both settled by agreement with measurement. In the narrow sense the equation has no adjustable parameter — given $\hbar$, m_e and

e the ground state follows. In the sense that matters for the word *exact*, it is a model of simple form whose form and whose scale were selected to fit.

And this is the point on which the whole question turns, so it is worth stating as a principle:

> *A fitted model is exact in the aspect it was fitted to.*

That exactness is automatic. It carries no information, because it was arranged. The informative question is never how well a model reproduces what selected it, but what it does *outside* the fit — and the answer there is a prediction, which can fail.

For hydrogen the fitted aspect is the bound-state spectrum: the $1/n^2$ pattern and, through h, its scale. Exactness there is free. The first genuinely unfitted predictions are the ones named in Section 1 — the fine structure and the $2s$–$2p$ splitting — and the equation gets both wrong, the second qualitatively, requiring Dirac and then quantum electrodynamics to repair. That is the ordinary and respectable fate of a good model tested beyond its fit. It is not the record of an exact equation.

3 The step that is not justified

For N electrons the equation is written in $3N$ dimensions,

$$\Big[-\tfrac{1}{2}\sum_i \nabla_i^2 - \sum_i \frac{Z}{r_i} + \sum_{i<j} \frac{1}{r_{ij}} \Big] \Psi(x_1, \ldots, x_N) = E\,\Psi, \tag{3}$$

with Ψ antisymmetric. This is a separate postulate. It is not derived from (1); it is (1) generalised by writing the same two ingredients once per electron and adding pairwise repulsion, and it carries two further components — antisymmetry and spin — that one-electron hydrogen never required.

The generalisation is natural and may well be right. What is not available is evidence that it is *exact*, and the reason is that nobody has solved it. Direct solution of (3) costs exponentially in N and is out of reach beyond three or four electrons. Every number the standard formalism produces for a real molecule comes from an approximation — a basis set, a functional, an active space, a truncated expansion — and the accuracy one can state depends on which approximation was chosen rather than on the equation.

So the claim "exact in principle, merely uncomputable" has an unusual structure. It cannot be tested, because testing it would require the computation that is said to be impossible. And the inductive support offered for it is the hydrogen case — where, by Section 2, the exactness is in the fitted aspect and therefore empty, and where, by Section 1, the first unfitted predictions fail and the spectrum is qualitatively wrong about a degeneracy. An extrapolation from a case that does not hold to cases that cannot be checked is not a strong position, however widely it is held.

4 The showcase, and what it shows

The figure usually produced at this point is the electron's anomalous magnetic moment. Theory and experiment agree on $a_e = (g-2)/2$ to roughly twelve significant figures [3, 4], and this is regularly described as making quantum mechanics the most precisely tested theory in science. The description is fair; the inference drawn from it is not.

The quantity is one-particle. It concerns a single electron's interaction with the quantised electromagnetic vacuum, computed as a perturbative series in α within quantum electrodynamics. It involves no many-electron wavefunction, no antisymmetry among several electrons, no $3N$-dimensional equation. It is evidence — overwhelming evidence — that QED describes the vacuum coupling of one charge. It is not evidence about (3), and offering it in support of the many-electron equation confuses two different theories with one Lagrangian between them.

Two further points are worth making precisely. The comparison requires α, and the most precise determinations of α now come either from a_e itself or from atom-recoil measurements whose results have at times disagreed at the level of the comparison [5, 6]; so part of the agreement is internal rather than independent. And the precision is a property of the observable, not of the theory's reach: twelve digits on a vacuum quantity coexists with no ability to state an error bar on a hundred-electron molecule.

5 The accuracy that collapses

Set out the standard formalism's precision as a function of electron count, and the shape is the thing to notice.

system	electrons	standard QM	continuum model
H	1	analytic, 4 figures vs measurement	exact (-0.5 by construction)
He	2	$\sim$20 digits [7]	2–4%
H_2	2	$\sim$12 digits [8]	8.6% total, 2.3% bond length
atoms to Ar	2–18	$\sim$1 kcal/mol, heavy machinery	1.6–4.5%
protein	$\sim 10^4$	**no number**	few per cent, 153 residues

At two electrons the gap is some fifteen orders of magnitude in the standard formalism's favour. At ten thousand there is no comparison to make, because one column is empty. The twenty digits for helium are obtained by explicitly correlated expansions in interelectronic coordinates — methods that do not scale and are not truncations of any general procedure. They are evidence about the two-electron problem, which is specially tractable, and not about the theory's reach.

6 A different generalisation, with no claim to exactness

The alternative examined here keeps the two ingredients of (1) — Coulomb interaction and the kinetic form (2) — and generalises differently. Instead of one antisymmetric function of $3N$ variables, N charge densities occupy *non-overlapping domains* in ordinary three-dimensional space, meeting with continuity across free boundaries whose position the energy determines [9]. Exclusion is realised geometrically rather than by antisymmetry; there is no spin; Z is the only quantity supplied.

Nothing argues that this is exact, and we do not claim it. It is a postulate, chosen for the same kind of reason (2) was chosen: it is the simplest generalisation of the two ingredients that remains three-dimensional, and it can be computed. Its predictions are therefore to be checked rather than deduced, and the checking has been done over a wide range:

- total atomic energies within 1–2% on average across the first three periods, worst case 4.5% at silicon, every atom solved the same way with nothing fitted [10];

- helium at -2.845 against -2.904, 2.0%, on the spherically homogenised solver;
- H_2 at 8.6% on the total energy with the bond length at 1.368 against 1.401, 2.3%, from commensurate separations on a converged mesh;
- first ionization energies of the alkalis and second of the alkaline earths within 1.4% and 4% respectively, from one constant per column, against the measured radius of the ion that remains [11];
- the row lengths $2, 8, 8, 18, 18, 32, 32$ and closure at all seven noble gases, from the pair and the octet with no $2n^2$ entering;
- hydrogen-bond geometry and secondary structure in polypeptides of up to 153 residues, carried with their electrons on a laptop.

The failures should be stated with the same directness. The H_2 *binding* energy comes out 58% high — which is the 8.6% error on the total divided by the bond energy being a seventh of it, not an independent defect, but 58% nonetheless. Ionization energies require a radius supplied from measured ionic radii, so that result is a correlation between two measured quantities and not a prediction from Z. Proton affinities come out with the wrong sign. Dispersion-bound systems are outside range. The model is not accurate in the sense the standard formalism is accurate for helium, and it will not become so.

7 What the comparison is about

The two models are both postulates. Both generalise Coulomb interaction and a chosen kinetic form to many electrons, in different ways, and neither generalisation is derived. They differ in what can be done with them.

One is exact in a limit nobody can compute, is wrong at 5×10^{-4} in the one case where exactness is claimed, achieves extraordinary precision for one and two electrons by methods that do not scale, and offers no error bar at all past a few dozen. The other is exact nowhere, agrees with measurement to a few per cent, and does so at every size from one electron to ten thousand, in three dimensions, at linear cost, with nothing fitted.

The first is more accurate wherever it can be evaluated. The second is *answerable* wherever matter exists. A theory whose error is knowable and roughly constant can be held to account at every scale; a theory whose error is unknowable past four electrons can be held to account only there. Which property matters more depends on what one wants a model of matter for — and that is a question about scope, which is a legitimate scientific question, rather than a question about precision, which the twelve digits have already settled in a domain where nobody disputes them.

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